

Day 18 — Method Comparison

Original YOLOv8n vs. White Balance + CLAHE + YOLOv8n, evaluated separately on all three datasets

Completed Work

- Compared Original YOLOv8n (baseline) and WB+CLAHE + YOLOv8n (enhanced) methods
- Evaluated separately on DIATAquarium, Aquatic Plant and Well
- Compared Precision, Recall, mAP@0.5, mAP@0.5:0.95 and inference speed
- Included image-quality observations from brightness and contrast analysis
- Tracking not evaluated — no video/sequential data available

Main Results — Detection Metrics

Dataset	Method	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Time/img	FPS
DIATAquarium	Original + YOLOv8n	90.02%	92.09%	92.72%	65.90%	21.25 ms	47.05
DIATAquarium	WB+CLAHE + YOLOv8n	87.70%	90.20%	91.10%	64.00%	29.96 ms	33.38
Aquatic Plant	Original + YOLOv8n	94.64%	93.52%	98.05%	70.79%	12.60 ms	79.35
Aquatic Plant	WB+CLAHE + YOLOv8n	94.60%	97.90%	98.70%	70.70%	13.93 ms	71.78
Well	Original + YOLOv8n	35.60%	36.80%	36.50%	16.40%	18.55 ms	53.91
Well	WB+CLAHE + YOLOv8n	36.30%	37.70%	35.20%	15.70%	15.82 ms	63.23

Evaluation set: DIATAquarium = validation (no local test folder); Aquatic Plant = validation; Well = test set.

Limitations & Failure Patterns

Data constraints, overfitting consideration, complexity and dataset-specific failure modes

Data Limitations

- DIATAquarium: no separate local test set; evaluated on validation only
- DIATAquarium: original (30 epochs) and enhanced (20 epochs) runs are not epoch-matched — flagged as a limitation on this comparison
- Source-level overlap identified across supplied splits (all three datasets)
- Aquatic Plant: single class; large proportion of background/empty-label images
- Well: 4 classes, strong imbalance — fishes dominates; stone and Inlet-pipe have very few instances

Overfitting Discussion

- Potential overfitting was considered during evaluation
- Final validation/test metrics alone are not sufficient to conclusively establish overfitting
- Training vs. validation curves should be examined in deeper experiments to check for divergence

Tracking Consistency

- No video or sequential data available at this project stage
- MOTA / IDF1 tracking metrics not evaluated — consistent with mentor-confirmed scope

Failure Patterns by Dataset

- DIATAquarium: enhancement reduced all four detection metrics; Fishes class comparatively weaker; processing time increased
- Aquatic Plant: already strong on original images; enhancement mainly improved Recall (+4.38pp); mAP changes small
- Well: overall performance far below other datasets; rare classes hard to detect; stone was not detected in the reported test evaluation — driven by severe class imbalance

Complexity & Processing

- Same YOLOv8n architecture used for both methods — only the input pipeline differs
- Enhanced pipeline: Original Image → White Balance → CLAHE → YOLOv8n
- Processing-time effect was dataset-dependent, not uniform

Dataset	Original	Enhanced	Change
DIATAquarium	21.25 ms	29.96 ms	+8.71 ms
Aquatic Plant	12.60 ms	13.93 ms	+1.33 ms
Well	18.55 ms	15.82 ms	−2.73 ms

Shortlisted Candidates & Next Plan

Strongest candidates identified for deeper experiments

Candidate 1 — Aquatic Plant + WB+CLAHE

- Recall improved: 93.52% → 97.90%
- mAP@0.5 improved: 98.05% → 98.70%
- mAP@0.5:0.95 remained nearly unchanged
- Detection performance stayed high overall
- Next: confirm recall improvement is stable with further training/validation analysis

Candidate 2 — Well + WB+CLAHE

- Precision: 35.60% → 36.30%; Recall: 36.80% → 37.70%
- FPS improved: 53.91 → 63.23
- mAP@0.5 and mAP@0.5:0.95 decreased
- Useful as a challenging / stress-test case
- Next: investigate class imbalance, rare-class performance and augmentation

Day 18 Conclusion

WB+CLAHE showed different effects across the three datasets. Aquatic Plant improved, DIATAquarium decreased, and Well showed small mixed changes. Therefore, the effect of enhancement is dataset-dependent. Aquatic Plant and Well were selected for further tuning, with Original YOLOv8n as the baseline.

Baseline (Retained)

- Original YOLOv8n kept as the control / reference method for all future experiments

Future Enhancement Candidates

- Water-Net — identified in literature review, not yet implemented
- FUnIE-GAN — identified in literature review, not yet implemented

Next Plan

- Investigate the two shortlisted methods in depth
- Examine training/validation curves for possible overfitting
- Analyze class-wise failure patterns further
- Explore improvement strategies for the Well dataset (imbalance handling)
- Maintain comparable experimental configurations (equal epochs) across all comparisons